MIL-T-10629C
23 March 1978
SUPERSEDING
MIL-T-10629B
20 July 1960

MILITARY SPECIFICATION

T-SQUARES (STEEL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steel T-squares used in drafting, layout, and reproduction work.

1.2 Classification. T-squares shall be of the following types and sizes, as specified (see 6.2).

Type I - Fixed head.
Sizes - 18-, 24-, 30-, 36-, and 42-inches long.

Type II - Protractor head.
Sizes - 24- and 42-inches long.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-766	- Steel Plates, Sheets, and Strip Corrosion-Resisting.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The T-squares shall consist of a head and a blade.

3.2 First article (first produced T-squares). When specified (see 6.2), the contractor shall furnish one or more T-squares for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce T-squares that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Blades. The blades shall be of steel conforming to QQ-S-766, Class 302, Finish 6.

3.3.2 Heads. The heads of the T-squares shall be of aluminum, magnesium, or zinc alloy.

3.3.2.1 Locking device and pivot. The locking device and pivot of the Type II T-square head shall be of corrosion-resistant steel.

3.4 Design.

3.4.1 Blades. Both ruling edges of the blade shall be parallel to each other, and shall be straight to within 0.003 inch per foot in length. A 3/8- to 1/2-inch diameter hole shall be centered near the free end of the blade to permit hanging the T-square. The length of the blade, in inches, shall be permanently marked on the blade. The indicated length of the blade shall not include the portion overlapping the head. Blades for the 18- and 24-inch T-squares shall be between 1-1/4- and 1-1/2-inches wide and 0.042- to 0.080-inch thick. Blades for the 30-, 36-, and 42-inch T-squares shall be between 1-1/4- and 2-inches wide and 0.042- to 0.100-inch thick. Tolerances shall be plus or minus 1/8 inch for the width and plus or minus 0.005 inch for thickness.

3.4.2 Heads. The head shall be of either solid or openwork design, and shall be so weighted as to give proper balance when handled. The working edge shall be machined to a smooth surface. The length of the heads for the 18- and 24-inch T-squares shall be not less than 10 inches, and the length for the 30-, 36-, and 42-inch T-squares shall be not less than 12 inches.

3.4.2.1 Finish. The heads shall be either natural finish or black enameled. When aluminum is used, the heads shall be anodized in accordance with MIL-A-8625, type optional, undyed.

3.5 Identification marking. Each T-square shall be identified in accordance with MIL-STD-130.

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3.6 Type I, fixed head. The head and the blade shall be permanently fastened together at right angles to each other. The angle between the head and the blade shall be 90° plus or minus 5 minutes.

3.7 Type II, protractor head. The head shall be fastened to the blade by a pivot which permits the head to pivot at least 80° from either side of zero setting. Zero setting shall be when the blade is at right angles to the head. When the index mark on the blade is set at zero on the protractor, the angle between the head and the blade shall be 90° plus or minus 5 minutes. Graduations shall be at every degree of arc and shall have a vernier scale to permit readings to 5 minutes of arc. At least each 10° of arc shall be numbered. A knurled thumbscrew or other locking device shall be provided to lock the head positively in any position. When the locking device is loosened, the head shall pivot freely and easily without binding, rough spots, or excessive play.

3.7.1 Graduations and numbering. Graduations and numbering shall be cut into the head and shall be filled with a permanent black ink or enamel. Graduation marks shall be 0.005-inch wide plus or minus 0.001 inch. When the T-squares are tested as specified in 4.3.2.1, graduation marks shall be within 1 minute of their true position.

3.8 Workmanship. The quality of workmanship shall be such as to produce T-squares that are in accordance with the requirements of this specification and are so constructed as to insure proper functioning of all parts of the unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced T-squares inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced T-squares inspection.

4.3.1 Examination. The first produced T-squares shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced T-squares shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.5 percent defective.

4.5 Inspection procedures.

4.5.1 Examination. The T-squares shall be examined as specified herein for the following defects:

Major

101. Sizes not as specified.
102. Material not as specified.
103. Incorrectly assembled.
104. Components missing or not as specified.
105. Dimensions not as specified.
106. Finish not as specified.
107. Workmanship not as specified.

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Minor

201. Identification marking not as specified.

4.5.2 Tests.

4.5.2.1 Perpendicularity and angular measurements. The angle between the blade of Type I T-squares and the contact edge of the head shall be measured with a protractor of known accuracy. Evidence of the angle between the head and the blade not 90° plus or minus 5 minutes shall constitute failure of this test. Type II T-squares shall be set at zero and also at angles of 15°, 30°, 45°, 60°, and 75° on each side of zero, and the angles shall be measured with a protractor of known accuracy. Evidence of the angle between the head and the blade not within plus or minus 5 minutes of the angles specified herein shall constitute failure of this test. When tested in an ambient temperature of 75° F (24° C) evidence of the Type II T-square graduation marks not within 1 minute of their true position shall constitute failure of this test.

4.5.2.2 Blade straightness test. Each ruling edge of the T-square shall be placed against a true straightedge, machinist's flat, or gage block, and the deviation from the reference surface checked with a feeler gage. Evidence of the T-square not straight within 0.003 inch per foot in length shall constitute failure of this test.

4.5.2.3 Parallelism test. The width of the blade shall be measured 2 inches from each end. Divergence of the two readings by more than 0.005 inch shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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108. Materials, method, or containers not as specified. Each incorrect material, method, or container shall constitute one defect.
109. Consolidation containers not closed Method V as specified.
110. T-squares of unlike description preserved or packed together.
111. Strapping not as specified.
112. Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each T-square shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.1.1 Consolidation. Preserved T-squares of like description shall be consolidated in close-fitting boxes conforming to PPP-B-636, W6c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. T-squares shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. T-squares of like description preserved as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2) strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. T-squares shall be packed as specified in 5.2.1 for Level A except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. T-squares preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The T-squares are used in general drafting practice and layout work where a heavy and rigid T-square is desired.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of T-square required (see 1.2).
(c) Time frame required for submission of first produced T-squares and number of units when required (see 3.2).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When Finish A strapping is required (see 5.2.1).

6.3 First produced T-squares. Any changes or deviations of production T-squares from the approved first produced T-squares during production will be subject to the approval of the contracting officer. Approval of the first produced T-squares will not relieve the contractor of his obligation to furnish T-squares conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification.

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Air Force - 99

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MIL-T-10629C T-Squares (Steel)

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